# Six clues inside a zipper

How a Zipper Holds Shut and Comes Apart

Name _____ Date _____



Big and Little found a jacket hanging on a peg, its zipper half closed. Above the slider the two sides were joined in one neat row. Below it they hung apart. Little wondered how one small piece could close a whole jacket. Big said a zipper is a team of small parts, each with its own job.



Laid flat on a table, a zipper shows its three main parts. The tape is the cloth strip that gets sewn to the jacket. The teeth, also called elements, are fixed in a row along each tape. The slider rides along both rows. Pull it one way and it joins the teeth. Pull it the other way and it parts them.



Up close, each tooth has a little bump on one side and a hollow on the other. The left teeth sit halfway between the right teeth, like fingers ready to fold together. Inside the slider, the two rows are squeezed together at an angle, so each bump tucks into the hollow of the next tooth. Teeth that fit together like this mesh, and the joined row is called the chain.

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|---|---|
| **tape** | the cloth strip of a zipper that gets sewn to a jacket or bag |
| **elements** | another name for a zipper's teeth |
| **slider** | the piece you pull to join or part the teeth |
| **mesh** | to fit together closely, like zipper teeth do |
| **chain** | the joined row of zipper teeth |
| **retainer box** | the small box at the bottom of a jacket zipper that holds the pin |

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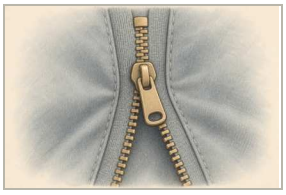
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Little tugged the two sides of the jacket, but the slider did not move. Big pointed under the slider. Many sliders hide a lock. When the pull-tab lies flat or is let go, a tiny piece slips between two teeth and holds the slider still. Lift the pull-tab and the piece moves out of the way, so the slider can glide again.



A jacket zipper comes all the way apart. It is called an open-end zipper. At the bottom, one tape ends in a thin insert pin, and the other ends in a small retainer box where the slider waits. The pin must go fully into the box. Then both rows start lined up, and the slider can mesh the very first pair of teeth.



At the very top of the zipper, Little found a tiny metal piece above the last teeth. It is a top stop. A stop at each end keeps the slider from running off the teeth. On a jacket, the retainer box is the stop at the bottom. Tape, teeth, slider, lock, pin, box and stops all work together each time a jacket closes.

1. What are the three main parts of a zipper, and what job does each one do?
2. How do the teeth mesh, and why doesn't the slider slide open on its own?
3. Why must the pin go fully into the box, and what do the stops do?

## Your turn

Write one sentence that tells how the slider closes a zipper.

Example: The slider squeezes the two rows, so each tooth tucks into the next one.